

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	RUTHIKA REDDY C	Reg. No.:	192421138
Section / Batch:	BTECH IT	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				

Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ASSESSMENT-CO3/AT2-CASE STUDY[← Back](#)Language: Python C C++ Java**QUESTIONS****Q1**

Digital Dictionary Lookup

**Q1****Digital Dictionary Lookup System
(Binary Search)**100
marks**SIMATS**
ENGINEERING**SIMATS Online Programming Lab****DAA PROGRAMMING****C RUTHIKA REDDY**
192421138RA**Your Code**

```
1 def binary_search_dictionary(dictionary, t
2     low = 0
3     high = len(dictionary) - 1
4
5     while low <= high:
6         mid = (low + high) // 2
7         mid_word = dictionary[mid]
8
9         if mid_word == target_word:
10             return f"Word '{target_word}'
11         elif mid_word < target_word:
12             low = mid + 1
13         else:
14             high = mid - 1
15
```

```
16     return "Word not found in the dictionary"
17 digital_dictionary = ["apple", "apply", "cat", "dog", "elephant"]
18 target = "cat"
19
20 result = binary_search_dictionary(digital_dictionary, target)
21 print(result)
```



Input

stdin input



Output

Word 'cat' found at index 4.

✓ Submitted